NAME: Ayush Mistry

ASSIGNMENT NO. 2
Enumeration

AIM: To conduct a comprehensive network enumeration of the target domain to identify and assess its security posture. This involves systematically discovering active hosts, open ports, and running services like Nmap, smbclient, enum4linux, theHarvester, Nessus and to gather information about potential security weaknesses and suggest ways to make system more secure.

Target: Metasploitable 2 (192.168.1.8)

TOOL USED: Nmap, smbclient, enum4linux, theHarvester, Nessus

TASKS:

1. Network Discovery:

Command: ping www.certifiedhacker.com

For this assignment, the target lab selected is **Metasploitable 2**. The target IP address for this network is **192.168.1.8**. This domain is used within authorized ethical hacking labs, making it suitable for practicing network enumeration techniques. This was done by pinging the domain to verify connectivity and resolve its IP address, ensuring the target is active and reachable before proceeding with further scanning and enumeration activities.

```
┌──(kali㉿kali)-[~]
└─$ ping 192.168.1.8
PING 192.168.1.8 (192.168.1.8) 56(84) bytes of data.
64 bytes from 192.168.1.8: icmp_seq=1 ttl=64 time=64.3 ms
64 bytes from 192.168.1.8: icmp_seq=2 ttl=64 time=1.22 ms
64 bytes from 192.168.1.8: icmp_seq=3 ttl=64 time=19.5 ms
64 bytes from 192.168.1.8: icmp_seq=4 ttl=64 time=28.3 ms
64 bytes from 192.168.1.8: icmp_seq=5 ttl=64 time=69.5 ms
64 bytes from 192.168.1.8: icmp_seq=6 ttl=64 time=20.0 ms
^C
--- 192.168.1.8 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5128ms
rtt min/avg/max/mdev = 1.217/33.811/69.512/24.816 ms
```

2. Scanning with Nmap:

Command: nmap -sV -O 192.168.1.8

- -sV: Service versions
- -O: Operating system

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV -O 192.168.1.8
Starting Nmap 7.95 ( https://nmap.org ) at 2025-09-09 19:17 IST
Nmap scan report for 192.168.1.8 (192.168.1.8)
Host is up (0.018s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec?
513/tcp  open  login
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:FC:9B:93 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux
:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 89.03 seconds
```

Key Data Extracted:

- **Host Status:** Up (0.018s latency)
- **Closed Ports:** 977 Closed ports
- **Operating System:** Linux 2.6.x (Ubuntu-based)
- SMB service is active on ports 139 and 445, running Samba 3.X–4.X
- Multiple vulnerable services are present (e.g., FTP, Telnet, Bindshell, VNC)

3. SMB Enumeration with smbclient and enum4linux:

Command 1: enum4linux -a 192.168.1.8

- -a: all simple enumeration

```
┌──(kali㉿kali)-[~]
└─$ enum4linux -a 192.168.1.8
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Tue Sep  9 20:07:45 2025

 =========================================( Target Information )=========================================

Target ........... 192.168.1.8
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none


 ============================( Enumerating Workgroup/Domain on 192.168.1.8 )============================


[+] Got domain/workgroup name: WORKGROUP


 ================================( Nbtstat Information for 192.168.1.8 )================================

Looking up status of 192.168.1.8
        METASPLOITABLE  <00> -         B <ACTIVE>  Workstation Service
        METASPLOITABLE  <03> -         B <ACTIVE>  Messenger Service
        METASPLOITABLE  <20> -         B <ACTIVE>  File Server Service
        ..__MSBROWSE__. <01> - <GROUP> B <ACTIVE>  Master Browser
        WORKGROUP       <00> - <GROUP> B <ACTIVE>  Domain/Workgroup Name
        WORKGROUP       <1d> -         B <ACTIVE>  Master Browser
        WORKGROUP       <1e> - <GROUP> B <ACTIVE>  Browser Service Elections

        MAC Address = 00-00-00-00-00-00

 ====================================( Session Check on 192.168.1.8 )====================================


[+] Server 192.168.1.8 allows sessions using username '', password ''


 =================================( Getting domain SID for 192.168.1.8 )=================================
```

```
 =================================( Getting domain SID for 192.168.1.8 )=================================

Domain Name: WORKGROUP
Domain Sid: (NULL SID)

[+] Can't determine if host is part of domain or part of a workgroup


 ===================================( OS information on 192.168.1.8 )===================================


[E] Can't get OS info with smbclient


[+] Got OS info for 192.168.1.8 from srvinfo:
        METASPLOITABLE Wk Sv PrQ Unx NT SNT metasploitable server (Samba 3.0.20-Debian)
        platform_id     :       500
        os version      :       4.9
        server type     :       0x9a03


 ========================================( Users on 192.168.1.8 )========================================

index: 0x1 RID: 0x3f2 acb: 0x00000011 Account: games    Name: games     Desc: (null)
index: 0x2 RID: 0x1f5 acb: 0x00000011 Account: nobody   Name: nobody    Desc: (null)
index: 0x3 RID: 0x4ba acb: 0x00000011 Account: bind     Name: (null)    Desc: (null)
index: 0x4 RID: 0x402 acb: 0x00000011 Account: proxy    Name: proxy     Desc: (null)
index: 0x5 RID: 0x4b4 acb: 0x00000011 Account: syslog   Name: (null)    Desc: (null)
index: 0x6 RID: 0xbba acb: 0x00000010 Account: user     Name: just a user,111,, Desc: (null)
index: 0x7 RID: 0x42a acb: 0x00000011 Account: www-data Name: www-data  Desc: (null)
index: 0x8 RID: 0x3e8 acb: 0x00000011 Account: root     Name: root      Desc: (null)
index: 0x9 RID: 0x3fa acb: 0x00000011 Account: news     Name: news      Desc: (null)
index: 0xa RID: 0x4c0 acb: 0x00000011 Account: postgres Name: PostgreSQL administrator,,,       Desc: (null)
index: 0xb RID: 0x3ec acb: 0x00000011 Account: bin      Name: bin       Desc: (null)
index: 0xc RID: 0x3f8 acb: 0x00000011 Account: mail     Name: mail      Desc: (null)
index: 0xd RID: 0x4c6 acb: 0x00000011 Account: distccd  Name: (null)    Desc: (null)
index: 0xe RID: 0x4ca acb: 0x00000011 Account: proftpd  Name: (null)    Desc: (null)
index: 0xf RID: 0x4b2 acb: 0x00000011 Account: dhcp     Name: (null)    Desc: (null)
index: 0x10 RID: 0x3ea acb: 0x00000011 Account: daemon  Name: daemon    Desc: (null)
index: 0x11 RID: 0x4b8 acb: 0x00000011 Account: sshd    Name: (null)    Desc: (null)
```

```
index: 0×12 RID: 0×3f4 acb: 0×00000011 Account: man     Name: man       Desc: (null)
index: 0×13 RID: 0×3f6 acb: 0×00000011 Account: lp      Name: lp        Desc: (null)
index: 0×14 RID: 0×4c2 acb: 0×00000011 Account: mysql   Name: MySQL Server,,,   Desc: (null)
index: 0×15 RID: 0×43a acb: 0×00000011 Account: gnats   Name: Gnats Bug-Reporting System (admin)       Desc: (null)
index: 0×16 RID: 0×4b0 acb: 0×00000011 Account: libuuid Name: (null)    Desc: (null)
index: 0×17 RID: 0×42c acb: 0×00000011 Account: backup  Name: backup    Desc: (null)
index: 0×18 RID: 0×bb8 acb: 0×00000010 Account: msfadmin        Name: msfadmin,,,       Desc: (null)
index: 0×19 RID: 0×4c8 acb: 0×00000011 Account: telnetd Name: (null)    Desc: (null)
index: 0×1a RID: 0×3ee acb: 0×00000011 Account: sys     Name: sys       Desc: (null)
index: 0×1b RID: 0×4b6 acb: 0×00000011 Account: klog    Name: (null)    Desc: (null)
index: 0×1c RID: 0×4bc acb: 0×00000011 Account: postfix Name: (null)    Desc: (null)
index: 0×1d RID: 0×bbc acb: 0×00000011 Account: service Name: ,,,       Desc: (null)
index: 0×1e RID: 0×434 acb: 0×00000011 Account: list    Name: Mailing List Manager      Desc: (null)
index: 0×1f RID: 0×436 acb: 0×00000011 Account: irc     Name: ircd      Desc: (null)
index: 0×20 RID: 0×4be acb: 0×00000011 Account: ftp     Name: (null)    Desc: (null)
index: 0×21 RID: 0×4c4 acb: 0×00000011 Account: tomcat55        Name: (null)    Desc: (null)
index: 0×22 RID: 0×3f0 acb: 0×00000011 Account: sync    Name: sync      Desc: (null)
index: 0×23 RID: 0×3fc acb: 0×00000011 Account: uucp    Name: uucp      Desc: (null)

user:[games] rid:[0×3f2]
user:[nobody] rid:[0×1f5]
user:[bind] rid:[0×4ba]
user:[proxy] rid:[0×402]
user:[syslog] rid:[0×4b4]
user:[user] rid:[0×bba]
user:[www-data] rid:[0×42a]
user:[root] rid:[0×3e8]
user:[news] rid:[0×3fa]
user:[postgres] rid:[0×4c0]
user:[bin] rid:[0×3ec]
user:[mail] rid:[0×3f8]
user:[distccd] rid:[0×4c6]
user:[proftpd] rid:[0×4ca]
user:[dhcp] rid:[0×4b2]
user:[daemon] rid:[0×3ea]
user:[sshd] rid:[0×4b8]
user:[man] rid:[0×3f4]
user:[lp] rid:[0×3f6]
user:[mysql] rid:[0×4c2]
```

```
user:[gnats] rid:[0×43a]
user:[libuuid] rid:[0×4b0]
user:[backup] rid:[0×42c]
user:[msfadmin] rid:[0×bb8]
user:[telnetd] rid:[0×4c8]
user:[sys] rid:[0×3ee]
user:[klog] rid:[0×4b6]
user:[postfix] rid:[0×4bc]
user:[service] rid:[0×bbc]
user:[list] rid:[0×434]
user:[irc] rid:[0×436]
user:[ftp] rid:[0×4be]
user:[tomcat55] rid:[0×4c4]
user:[sync] rid:[0×3f0]
user:[uucp] rid:[0×3fc]

 ==================================( Share Enumeration on 192.168.1.8 )==================================

        Sharename       Type      Comment
        ---------       ----      -------
        print$          Disk      Printer Drivers
        tmp             Disk      oh noes!
        opt             Disk
        IPC$            IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
        ADMIN$          IPC       IPC Service (metasploitable server (Samba 3.0.20-Debian))
Reconnecting with SMB1 for workgroup listing.

        Server               Comment
        ---------            -------

        Workgroup            Master
        ---------            -------
        WORKGROUP            METASPLOITABLE

[+] Attempting to map shares on 192.168.1.8

//192.168.1.8/print$    Mapping: DENIED Listing: N/A Writing: N/A
//192.168.1.8/tmp       Mapping: OK Listing: OK Writing: N/A
//192.168.1.8/opt       Mapping: DENIED Listing: N/A Writing: N/A
```

```
[E] Can't understand response:

NT_STATUS_NETWORK_ACCESS_DENIED listing \*
//192.168.1.8/IPC$      Mapping: N/A Listing: N/A Writing: N/A
//192.168.1.8/ADMIN$    Mapping: DENIED Listing: N/A Writing: N/A

 ===========================( Password Policy Information for 192.168.1.8 )===========================


[+] Attaching to 192.168.1.8 using a NULL share

[+] Trying protocol 139/SMB ...

[+] Found domain(s):

        [+] METASPLOITABLE
        [+] Builtin

[+] Password Info for Domain: METASPLOITABLE

        [+] Minimum password length: 5
        [+] Password history length: None
        [+] Maximum password age: Not Set
        [+] Password Complexity Flags: 000000

                [+] Domain Refuse Password Change: 0
                [+] Domain Password Store Cleartext: 0
                [+] Domain Password Lockout Admins: 0
                [+] Domain Password No Clear Change: 0
                [+] Domain Password No Anon Change: 0
                [+] Domain Password Complex: 0

        [+] Minimum password age: None
        [+] Reset Account Lockout Counter: 30 minutes
        [+] Locked Account Duration: 30 minutes
        [+] Account Lockout Threshold: None
        [+] Forced Log off Time: Not Set
```

```
[+] Retieved partial password policy with rpcclient:


Password Complexity: Disabled
Minimum Password Length: 0


 ======================================( Groups on 192.168.1.8 )======================================


[+] Getting builtin groups:


[+]  Getting builtin group memberships:


[+]  Getting local groups:


[+]  Getting local group memberships:


[+]  Getting domain groups:


[+]  Getting domain group memberships:


 ==================( Users on 192.168.1.8 via RID cycling (RIDS: 500-550,1000-1050) )==================


[I] Found new SID:
S-1-5-21-1042354039-2475377354-766472396

[+] Enumerating users using SID S-1-5-21-1042354039-2475377354-766472396 and logon username '', password ''

S-1-5-21-1042354039-2475377354-766472396-500 METASPLOITABLE\Administrator (Local User)
S-1-5-21-1042354039-2475377354-766472396-501 METASPLOITABLE\nobody (Local User)
S-1-5-21-1042354039-2475377354-766472396-512 METASPLOITABLE\Domain Admins (Domain Group)
```

```
S-1-5-21-1042354039-2475377354-766472396-513 METASPLOITABLE\Domain Users (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-514 METASPLOITABLE\Domain Guests (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1000 METASPLOITABLE\root (Local User)
S-1-5-21-1042354039-2475377354-766472396-1001 METASPLOITABLE\root (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1002 METASPLOITABLE\daemon (Local User)
S-1-5-21-1042354039-2475377354-766472396-1003 METASPLOITABLE\daemon (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1004 METASPLOITABLE\bin (Local User)
S-1-5-21-1042354039-2475377354-766472396-1005 METASPLOITABLE\bin (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1006 METASPLOITABLE\sys (Local User)
S-1-5-21-1042354039-2475377354-766472396-1007 METASPLOITABLE\sys (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1008 METASPLOITABLE\sync (Local User)
S-1-5-21-1042354039-2475377354-766472396-1009 METASPLOITABLE\adm (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1010 METASPLOITABLE\games (Local User)
S-1-5-21-1042354039-2475377354-766472396-1011 METASPLOITABLE\tty (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1012 METASPLOITABLE\man (Local User)
S-1-5-21-1042354039-2475377354-766472396-1013 METASPLOITABLE\disk (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1014 METASPLOITABLE\lp (Local User)
S-1-5-21-1042354039-2475377354-766472396-1015 METASPLOITABLE\lp (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1016 METASPLOITABLE\mail (Local User)
S-1-5-21-1042354039-2475377354-766472396-1017 METASPLOITABLE\mail (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1018 METASPLOITABLE\news (Local User)
S-1-5-21-1042354039-2475377354-766472396-1019 METASPLOITABLE\news (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1020 METASPLOITABLE\uucp (Local User)
S-1-5-21-1042354039-2475377354-766472396-1021 METASPLOITABLE\uucp (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1025 METASPLOITABLE\man (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1026 METASPLOITABLE\proxy (Local User)
S-1-5-21-1042354039-2475377354-766472396-1027 METASPLOITABLE\proxy (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1031 METASPLOITABLE\kmem (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1041 METASPLOITABLE\dialout (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1043 METASPLOITABLE\fax (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1045 METASPLOITABLE\voice (Domain Group)
S-1-5-21-1042354039-2475377354-766472396-1049 METASPLOITABLE\cdrom (Domain Group)

 ================================( Getting printer info for 192.168.1.8 )================================

No printers returned.


enum4linux complete on Tue Sep  9 20:08:36 2025
```

- Workgroup & Domain
- Workgroup Name: WORKGROUP
- Domain Name: METASPLOITABLE
- Server Type: Samba 3.0.20-Debian (Linux)
- **Built-in Groups:**
  Domain Users
  Domain Guests
  Domain Admins
  Local Users
  Local Administrators
- **Local Groups:**
  games, proxy, syslog, news, mail, daemon, etc.
- **Domain Groups:**
  Domain Users, Domain Guests, Domain Admins, Backup Operators, Print Operators, Remote Desktop Users, Dialout, Fax, Voice, cdrom
- **Security Identifier (SID):**
  Domain SID: S-1-5-21-1042354039-2475377354-766472396
  Used to uniquely identify the domain in Windows environments.

Command 2: smbclient //192.168.1.8/tmp

- tmp: Temporary Share is a publicly accessible SMB directory typically used for temporary file storage, but often misconfigured to allow anonymous access.

The system accepted an empty password, confirming that anonymous login is allowed.
Once connected, ran the **ls** command to list the contents of the share:

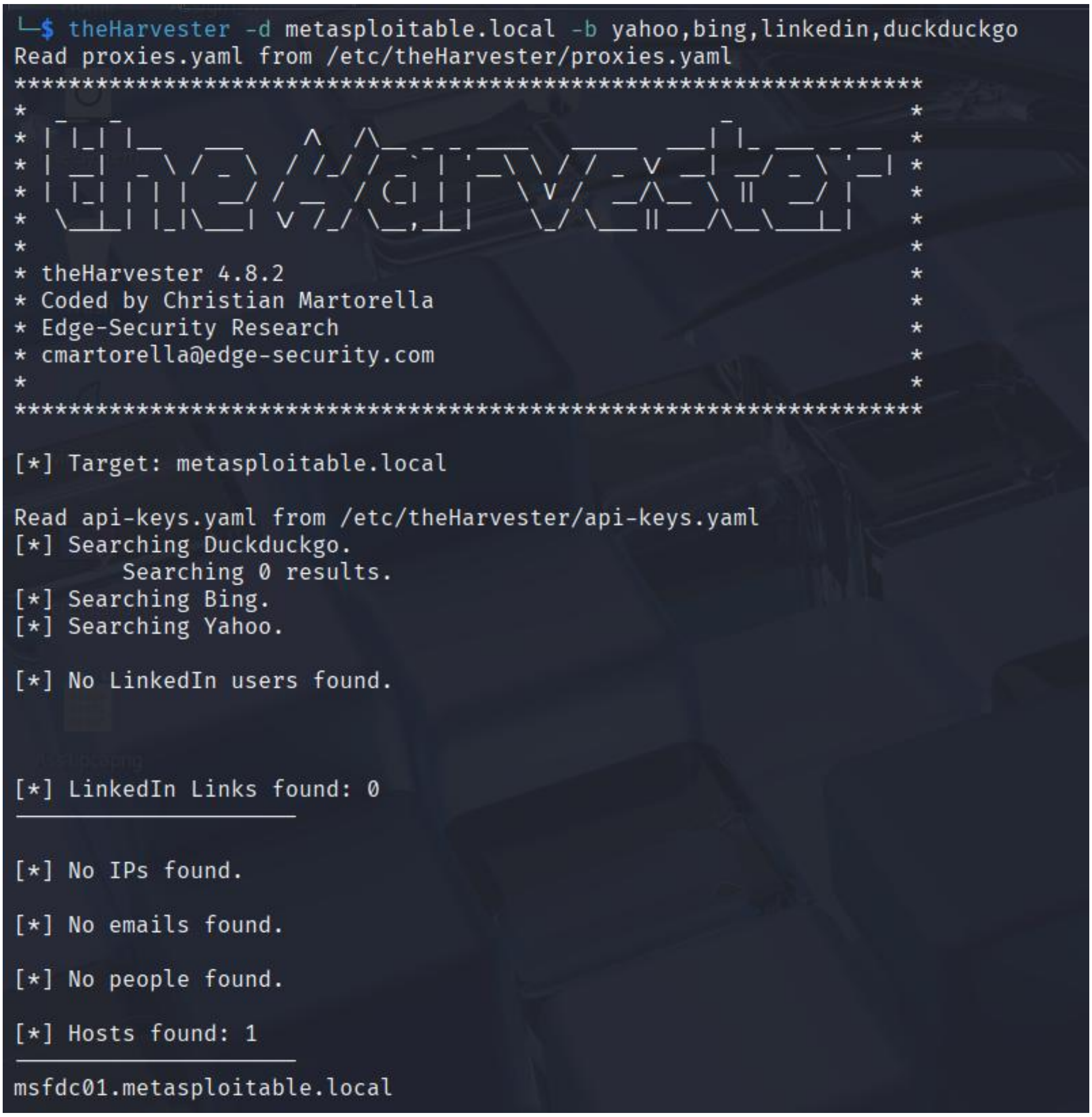
```
┌──(kali㉿kali)-[~]
└─$ smbclient //192.168.1.8/tmp
Password for [WORKGROUP\kali]:
Anonymous login successful
Try "help" to get a list of possible commands.
smb: \> ls
  .                                   D        0  Tue Sep  9 19:29:51 2025
  ..                                 DR        0  Mon May 21 00:06:12 2012
  .ICE-unix                          DH        0  Tue Sep  9 19:04:03 2025
  .X11-unix                          DH        0  Tue Sep  9 19:05:13 2025
  .X0-lock                           HR       11  Tue Sep  9 19:05:13 2025
  4639.jsvc_up                        R        0  Tue Sep  9 19:05:24 2025

                7282168 blocks of size 1024. 5436232 blocks available
smb: \>
```

- The tmp share is readable without authentication
- Files such as .ICE-unix, .X11-unix, and 4639.jsvc_up are present
- An attacker could potentially upload malicious files or exploit misconfigurations

4. Email Enumeration with theHarvester:

Command: theHarvester -d Metasploitable.local -b yahoo,bing,linkedin

- -d metasploitable.local: Specifies the target domain (local hostname of Metasploitable 2)
- -b: search across multiple engines



- **Result:** The tool returned no email addresses, IPs, or user profiles. A single host (msfdc01.metasploitable.local)was identified, but no organizational data was found.
- **Reason:** This is expected because **metasploitable.local** is a **local lab machine** with no public internet presence or indexed web content. It does not have an active online footprint, so no emails or employee data are available through public sources.

5. Vulnerability Scanning with Nessus:

A Vulnerability scan was performed on the target system 192.168.1.8 (Metasploitable 2) using Nessus Essentials. The scan identified 120 total vulnerabilities, including 9 Critical and 6 High severity issues.

Firstly, downloading Nessus in kali Linux then open https://localhost:8834 in browser

Total Vulnerabilities Found: 120

Breakdown:

- Critical: 9
- High: 6

- Medium: 20
- Low: 8
- Info: 77
- **Critical Vulnerabilities:**

Plugin ID	Vulnerability Name	Port	Description
134862	Apache Tomcat AJP Connector Request Injection (Ghostcat)	8009	Allows attacker to read arbitrary files from Tomcat server
51988	Bind Shell Backdoor Detection	1244	Metasploitable root shell exposed - critical remote access
20007	SSL Version 2 and 3 Protocol Detection	443, 993, 995	Weak SSL protocols allow man-in-the-middle attacks
171340	Apache Tomcat EoL (<= 5.5.x)	8180	Outdated, unsupported version with known exploits
201352	Canonical Ubuntu Linux EoL (8.04.x)	—	Entire OS is end-of-life → no security patches
46882	UnrealIRCd Backdoor Detection	6667	Remote shell via IRC - infamous Metasploitable backdoor
61708	VNC Server ‘password’ Password	5900	VNC accessible with default password “password”

- **High Severity Vulnerabilities:**

Plugin ID	Vulnerability Name	Port	Description
136769	ISC BIND Service Downgrade/Reflected DoS	53	DNS server vulnerable to denial-of-service
42256	NFS Shares World Readable	2049	Anyone can read shared files - data leakage risk
90509	Samba Badlock Vulnerability	445	Allows remote code execution via SMB
10205	rlogin Service Detection	513	Unencrypted remote login - credentials exposed
10245	rsh Service Detection	514	Remote shell without authentication

- Key High Vulnerabilities:
  NFS World Readable Shares (Port 2049) - Data leakage risk
  rlogin/rsh Services (Ports 513, 514) - Unencrypted remote access

Conclusion:

This whole assignment I explored a vulnerable lab machine (Metasploitable 2) to see what an attacker could find and how we can stop them. Firstly, started by scanning it with Nmap and found tons of open ports like FTP, SMB, Telnet, and even VNC with the password Then dug deeper into the SMB service and found log in was possible without any username or password and look inside shared folders. Then tried finding company emails with theHarvester, but since it is just a lab machine, there were none. Finally, ran a full vulnerability scan with Nessus and found over 120 security holes, including 9 critical ones that could let an attacker take full control of the system in minutes. Things like old software, weak passwords, and unencrypted services made this machine an easy target. If companies don’t scan, patch, and lock things down, higher chances of getting hacked. This project really showed me how important it is to find these problems and fix them fast.